

CHAMBERS'S JOURNAL

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CONDUCTED BY WILLIAM AND ROBERT CHAMBERS.

No. 739.

SATURDAY, FEBRUARY 23, 1878.

PRICE 1½d.

A WORD ABOUT BIRD-KEEPING.

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and he did so with as great tenderness of feeling as the fragile nature of the article dealt in demanded. He had much to explain respecting the importation of song-birds, and the breeding of them in cages. But on neither of these points shall we say anything. What especially interested us were this intelligent dealer's observations on the proper method of keeping birds. Some folks, he said, have a notion that all you have to do is to buy a bird, put it into a cage, and give it food and water as directed. That is far from being enough. The habits of the animal must be studied. The climate of the room in which it lives, the amount of daylight it should enjoy, the atmosphere it breathes, its freedom from sudden alarms—all have to be thought of, if you wish the bird to be happy; and without that it has little chance of being a pleasant companion.

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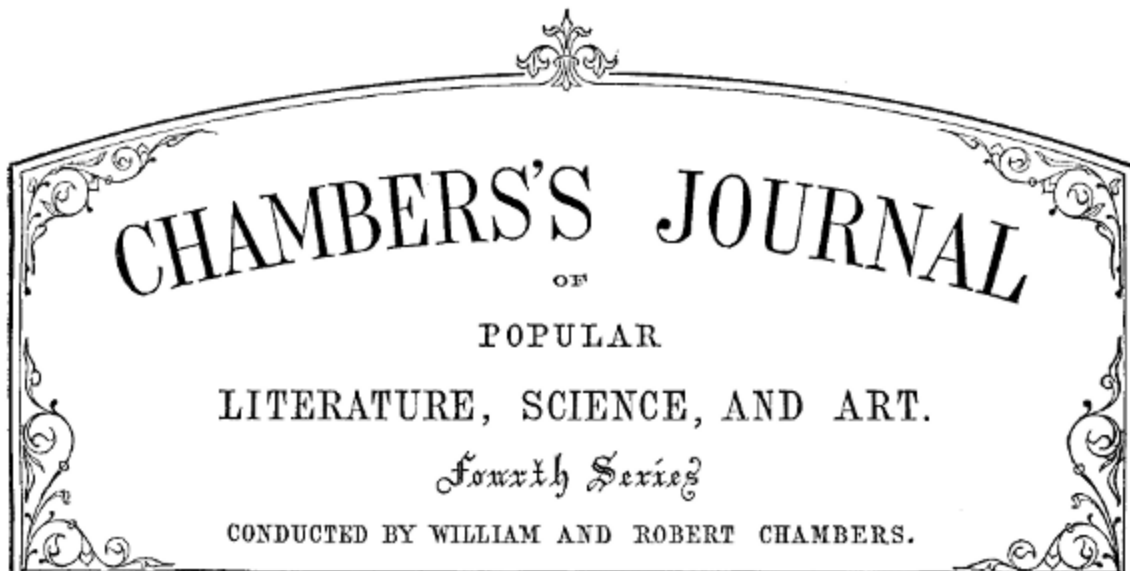
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What could be the matter? Puzzled to the last extent, the bird-dealer at length conjectured what might be the cause of these numerous deaths. Could it be that the birds wore themselves out singing? If so, the only way to stop them was to shorten the time they were exposed to the light, for if kept in the dark they are not inclined to sing.

The supposition proved to be correct. He shut up his shop at an early hour, and from that time the mortality of the birds ceased. During the day they had just that amount of singing that suited their constitutions, and in the evening they were left to their repose. This bird-dealer's ingenious discovery seems exceedingly rational. In a state of nature, small birds flit about and sing only during daylight. They retire to rest at sundown. This procedure requires to be imitated in keeping birds artificially. If you let them sing all day and several hours additional by lamp-light, you over-fatigue them. The labour is too much. Of course the birds do not understand that they had better be silent when the lamp or candles are lit. They instinctively keep singing on, as if it were still daylight. The immediate

effect of this over-fatigue is that the poor birds are apt to moult, and become attenuated; and suffering from premature exhaustion, they speedily perish.

The dealer mentions that few birds subject to the exhaustion of singing beyond ordinary daylight survive more than two years. This does not surprise us. How could any of our public vocalists, male or female, and of even a robust constitution, endure the tear and wear of singing under a mental strain for any great length of time, as much as eighteen hours a day? If human beings would thus sink under the effort of over-work, we need not wonder that the fragile creatures we are speaking of should succumb and drop from their perch.

As a means, therefore, of protecting the lives of pet birds, the recommendation is, to remove the cages to a darkened apartment at nightfall, or if they are not removed, to cover up every cage with a dark cloth before lighting the gas or oil-lamps. In shifting birds from one room to another, it is important to see that there be no change in the temperature. If removed to a different temperature, there is a chance of their moulting, which may be preliminary to something more serious. Let it be always kept in mind that Nature supplies a coat to suit the heat or cold in which the creatures are placed. By changing a bird from a warm to a cold climate, birds change their coat and get one that is heavier, and *vice versâ*, so, by repeated changes they are kept continually moulting, instead of once a year, as they ought to do.

We have referred principally to the treatment of small song-birds, the delicacy of which calls for particular attention. But our observations in the main apply to all birds whatsoever. If it be wrong to keep a little bird singing beyond its constitutional capacity, so it would be wrong to over-work a parrot by causing it to speak eighteen hours on a stretch. It would seem that by this degree of loquacity, the parrot has a tendency to take some kind of bronchial affection, analogous to the ailment of preachers, usually known as 'the minister's sore throat,' and which, if not checked in time, may prove equally disastrous.

We have thrown these interesting facts together not only in the interest of bird-keepers, but for the sake of inculcating kindness to animals.

W. C.

MY KITMITGHAR 'SAM.'

FOR nearly three years my Kitmitghar, as that functionary is called, was cook, butler, and factotum of my then small bachelor establishment in India. A cunning concocter of mulligatawnies, curries, and chutnies—as cunning a hand too in ‘cooking’ his daily bazaar accounts, adding annas and pice, for his own particular benefit, to the prime cost of as many articles as possible. Mildly remonstrated with, and petty larceny hinted at, his honest indignation would be aroused. ‘Master tink I cheat,’ he would say; ‘master can inquire bazaar-mans;’ well knowing, the rogue, the moral and almost physical impossibility of ‘master’—a swell in his way—going to the distant market in a broiling sun, and finding out the ruling prices of flesh and fowl.

This worthy, whose original cognomen of *Mootoosammy* was shortened into ‘Sam’ for convenience and euphony sakes, was a Tamil from the Malabar Coast. *Au reste*, a dark, handsome, stoutly-built, clean-looking native, on whose polished skin water and coarse country soap were evidently no strangers. In his early youth, fated to earn his own living, he had been ejected from the paternal hut and placed as a *chokerah* or dressing-boy to a fiery and impecunious lieutenant of infantry; and under the fostering care of that impetuous and coinless officer, his indoctrination into the art and mystery of a valet had been advanced and improved by sundry ‘lickings,’ and by frequent applications to his ebon person of boot-heels, backs of brushes, and heavy lexicons of the English and Hindustani languages. This education completed, and when he had learned to appreciate the difference between uniform and mufti, mess-dress and parade-dress, and indeed to master the intricacies of his employer’s scanty wardrobe—*non sine lacrymis*, not without ‘howls’—then he emerged from dressing-boyhood, was promoted *matie* or under-butler, and got translated into more pretentious bungalows than those of indigent subalterns. By-and-by further preferment awaited him; he became *kitmitghar* (major-domo) in the households of unmarried civilian or military swells, and thenceforward led a life free from kicks and cuffs, canes and whips, and impromptu missiles snatched from toilet or study tables. I have said advisedly ‘unmarried,’ for except under financial difficulties, Sam would not take

service with the Benedicts of Indian society, and the actual presence or possible advent of a wife was the signal for his departure. 'Plenty too much bodder wid lady; too much want ebery day, ebery day measure curry stuff, oil, ghee [butter]; too much make say always dis ting too dear, dat ting too dear; too much trouble take count. Now, Colonel Sahib he good man; he call, he say: "Sam! how much this week you eespend? [spend]." He just look book; he give rupee; no one single word *bobberee* [fuss] make.' And so, for a palpable reason, my worthy cook-butler eschewed those households where a better-half took the reckoning.

English, after the rickety fashion of a Madrassee, Sam spoke fairly enough; he also read and wrote the language, the latter accomplishment phonetically, but yet sufficiently near to the rules of orthography to make you fully understand and pay for 'tirty seers wrice' as thirty seers (measures) of rice. What if he did elect to spell rice with a w? Is it not recorded that an eminent member of a large mercantile firm, in days long gone by, invariably included an *h* in the word sugar? And is it not also chronicled how he chastised almost to the death his son and heir for omitting that letter when invoicing a cargo of best Jamaica moist? If then Blank Blank, Esq. of the city of London opined that sugar required an *h*, why not the same liberty as regards the *w* to Mootoosammy of the city of Madras?

A sad waverer in religious opinions Master Sam, I fear. A very Pharisee of a Hindu, a rigid stickler for the worship of Vishnu or Siva on the high-days and holidays of those deities, when his forehead and arms would be spotted and streaked with coloured ashes, his garments would smell of saffron and sandal-wood, his English diminutive name would be put aside for its more lengthy and sonorous native patronymic, and he would be off to the temple to make *poojah* (prayer) to his *swamis* (gods). But yet, somehow or other, all these symptoms and signs of Hinduism would disappear at Christmas, Easter, or Whitsuntide. At those seasons of the Christian year, Sam was no longer Mootoosammy, but Sam pure and simple. No more the believer he in the Vedahs and Shastras, but a pinner of faith on Aves and Credos; no *poojah* for him now in the temple, but crossings and genuflections in the little chapel of the station. Not a trace in these days of idolatrous scents clinging to cloths and turban, or of 'caste' marks disfiguring brow or limb. Dole in hand—obtained either from pickings at master's 'counts or from bazaar-man's *dustoor* (custom)—he is off to join Father Chasuble's small

flock, and to bow down and formalise with the best or worst of that good priest's congregation. I really think and believe, that to secure a holiday and an 'outing,' Sam would have professed himself a Mohammedan during the Ramadan, a Hebrew during the Passover, a Heathen Chineese during the feast of Lanterns, and a Buddhist during the Perihara or other high-jinks of the yellow-robed priests of Gautama Buddha.

I never before or since met any man into whose household death was so constantly making inroads, and strange to say, carrying away the same individual. I suppose that, on a rough estimate, all Sam's kith and kin died at least twice during the thirty months or so that he was in my service.

'Master please'—thus Sam howling and weeping after his kind—'scuse [excuse] me. Gib tree day leave go Madras; too much trouble my house. My poor old mudder—booh! ooh!—plenty long time sick; master know well; too much old got; die last night. Booh! ooh! o-o-g-h!'

'Why, what tomfoolery is this?' I reply. 'Your mother dead! Dead *again*! Why, man, how can that be? Four months ago you came and told me your mother was dead; you got four rupees advance; you went off, leaving the boy to do your work, and put me to no end of inconvenience. How can the old woman be dead again?'

But the fellow is not the least put out, and is quite equal to the 'fix.' 'Master Sahib,' he says, 'I beg you scuse me. Sahib quite wrong. That time you speak I get leave, not *my* mudder—my *wife's* mudder die. Master can look book!'

This random shot anent the 'book' alludes to my diary, in which the disbursement of the money has been entered, but not of course the casualty in his family. But I don't lose the hint nevertheless, and I jot down a memorandum for future reference, should occasion require.

Then Sam goes on: 'I no tell lie, sar. Plenty true; too much bobberee my house make. My fader gone Mysore'——

'Why, bless my heart!' I put in, 'you told me ages ago your father died of cholera in Masulipatam.'

'No, sar,' says Sam; 'never, sar! My grand-fader, scuse me. My wife she catch bad fever. No one single person my home got, make funeral-feast. Please, my master, advance half-month's pay; gib four days' leave. I too

much hurry come back.' Then he falls down, clasps my feet, calls me his father, brother; gets my consent to be absent, handles the rupees, and is off like a shot; not of course to his mother's obsequies, for the old harridan has either been buried or burned years ago, or even now is all alive and kicking; but to some spun-out native theatricals, nautch, or *tamasha* (entertainment) in Black Town, where he feasts, drinks, and sleeps, and for a week at least I see his face no more.

History repeats itself; so does Sam. Months and months have passed; I am away from the neighbourhood of the Presidency town, and on the cool Neilgherry Hills. Enters one morning my man into my sitting-room, a letter in his hand, written in Tamil, and which he asks me to read, well knowing that I can't, that except a very few of the commonest words of the language, which I speak with an uncertain not to say incorrect idea of their meaning, the tongue of his forebears, scriptural and oral, is to me Chaldee or Arabic.

'Well! what's up now?' I say '*Ennah?*' airing one of the expressions I know.

'Master can see self. My uncle he send chit [note]; just now tappal-man [postman] bring. He write, say: "Sam! you plenty quick come Madras." He put inside letter one five-rupee government note. Sahib can see. He tell me no one minute lose; take fire-road [railway]; too soon come; plenty, plenty trouble. *My mudder dead.*'

'You awful blackguard!' I exclaim. 'Your mother dead—dead again! Look here—look here!' And I turn up my diary and shew him, under date August 9, 186-, nearly two years past and gone: 'Sam's mother reported dead for the second time by Sam, &c.'

Then he slinks away discomfited; and I hear him in his smoky kitchen growling and grumbling, and no doubt anathematising me and mine past, present, and future.

My first introduction to Sam was after this wise. I had come down from Bombay to Beypore with troops in a small steamer, and Mr Sam, who had either deserted or been sent away from the Abyssinian Expedition, in which he had been a camp-follower, was also a passenger in the same ship. Of this craft a word *en passant*, for I have to this day a lively and by no means pleasant olfactory recollection of her. She was the dirtiest vessel in which I ever put foot; guiltless of paint from keel to truck; all grime, coal-soot, and

tar from stem to stern. She had but recently taken a cargo of mules to Annesley Bay; and but scant if any application of water and deodorants had followed the disembarkation of the animals. The 'muley' flavour still therefore clung closely to bulkhead and planking; it hung about cordage and canvas; it penetrated saloon and sleeping-berth; it even overpowered the smell of the rancid grease with which pistons and wheels were lubricated. Worthy Captain B—— the skipper assured us that deck and hold, sides and bulwarks, had been well scoured in Bombay; but as the old salt's views of scrubbing, judging from his personal appearance, were infinitesimally limited, we opined that the ship's ablution had been as little as was that of its commander's diurnal tub.

But to return to Sam. The poor fellow was wandering about the streets of Beypore coinless and curry-and-rice-less, when he stumbled upon me. He was seeking, he told me, from some good Samaritan of an officer, a free convoy to Madras as his servant; and as I happened to be in a position entitled to passes for some three or four followers at government expense, I was enabled to pour oil and wine into Sam's wounds, and without even the disbursement to mine host the assistant-quartermaster-general, of the traditional 'tuppence,' to get him across from terminus to terminus—some four hundred long miles—and without once casting eyes on him. But at Lucifer's hotel in Madras where I stayed—— What a memory of mosquitoes, fleas, and other nimble insects doth it bring! What a night-band of croaking frogs and howling jackals it kept! What packs of prowling pariah dogs and daringly thieving crows congregated about its yards and outhouses! What repulsive nude mendicants and fakeers strolled almost into its very verandahs! What a staff of lazy sweepers, slow-footed 'boys,' and sleepy punkah-pullers crawled about it generally! And last, though not least, what a wretched 'coolie-cook' superintended its flesh-pots, from which not even the every-day stereotyped prawn curry, boiled seer-fish, and grilled *morghee* (fowl) could creditably and palatably issue. At this Stygian caravanserai then, Sam, whom I thought I had bid adieu to for ever and a day on the railway platform, turns up again clean and smirk, salaams, asks for permanent employment, produces a thick packet of highly laudatory characters (mostly, I had no doubt, either fabricated by a native scribe in the Thieves' Bazaar at Black Town, or borrowed for the occasion from some other brother-butler), gets engaged; and from that moment, both figuratively and literally, begins to eat my salt. Nor did the saline feasting fail to give

him a taste for liquor—for alcoholic, decidedly alcoholic were Sam's proclivities. He drank at all times and in all places; but his favourite day and locality was Tuesday, at the weekly market of the cantonment. Then and there he imbibed right royally, and staggering home—the coolies with the supplies following him as tipsy as himself—went straight to his mat-spread *charpoy* (bedstead).

'Hollo, Sam!' I exclaim; 'at it again; drunk as usual from *shandy* [market].'

'No, shar! Dis time no shrunk! Shun too mush hot! Splenshy head pain gib! Too mush make shake, sthagger, shar! No, mash-err, no! Sham not shrunk! Plenty shick! Shmall glass brandy—all right, shar!'

But I decline to add 'the sum of more to that which hath too much,' and I leave Sam to sober himself as he best can, and which, truth to say, he quickly does.

In the way of intoxicants nothing came amiss to my man's unfastidious palate. He had no particular 'wanity,' like Old Weller's friend the red-nosed Shepherd: Henneysey's brandy, Kinahan's whisky, Boord's gin, Bass's ale, Guinness's stout, champagne, sherry, claret—all and each were equally acceptable; and failing these European liquors, then the vile palm-toddy and killing mango-spirit of the neighbouring native stills supplied their place. Bar the toddy and mango stuff, which were cheap and easily obtained, Sam did not disburse much for his wine-cellar; master's sideboard and stores, guard them as he would, came cheaper and handier. Every bottle, somehow or other, got 'other lips' than mine and my friends' applied to it, and its contents went into and warmed other 'hollow hearts' than ours. Sam laid an embargo on and helped himself from all. He it is, I fancy, to whom Aliph Cheem alludes in his Lay of Ind entitled *The Faithful Abboo*, that trusty servant who, habitually stealing his master's liquor, and accusing his brother-domestics, got caught and half-poisoned by mistaking in his prowls Kerosine for Old Tom. A misadventure not unlike befell Sam; but in that instance he did not 'strike oil,' but came upon a very nauseating dose of tartar emetic, and was 'plenty sick' and 'plenty shame' for some hours after.

Another predilection of my factotum's was tobacco, which he smoked without ceasing, and without the least regard to quality or fabric. 'Long-cut or short-cut' to him 'were all the same.' But as I did not happen to be addicted to the 'nicotian weed' Sam could not draw on any resources of

mine, but had to depend on his own means, supplemented by the surreptitious abstraction of Trichys and Manillas, of Latakia and Bird's-eye, from the boxes and pouches of my chum and visitors.

Every native gambles; so it could hardly be expected that Sam should differ from his brethren in this respect. In the words of the old ditty anent Ally Croker:

He'd game till he lost the coat from his shoulder.

I don't think he cared much for cards or dice; but the game that he delighted in was played with a red and white checkered square of cloth, and with round pieces like draughtsmen. Whenever the advent of a friend and opportunity served, down the two squatted with this board between their legs, and a pile of copper pieces of money by their sides; and so intent would they be on their play, that nothing short of a gentle kick, or tap on the head, would arouse them to master's wants and needings.

My readers will naturally inquire why, with all these delinquencies, Sam so long remained my henchman. Well, first, had I discharged him, another and probably greater robber would have stepped into his shoes, and bazaar accounts and inroads on alcohol and tobacco would have remained undiminished. 'They all do it;' so better the de'il I knew, than the de'il whose acquaintance I would have to make. Again, Sam had his redeeming points; he was, as I have said before, clean, handy, and deft at the creature comforts, which, having appetisingly compounded, he could serve up with taste and elegance. Then he was a good nurse; and during a serious illness that befell me at one of the vilest stations in Madras, he tended me closely and carefully, keeping a watchful eye and a ready stick on punkah-pullers and wetters of kus-kus tatties (scented grass mats), without the cooling aid of which the heat of that grilling July would have been my death on that fever-bed. Once more, on those military inspections which fell to my lot, and which had to be undertaken partly over the Nizam's very sandy and rough highways, and in those close comfortless bone-breaking vehicles called *byle-nibbs* (bullock-carts), my man became invaluable. Seated on the narrow perch alongside the almost garmentless and highly odoriferous native driver, he urged him on by promises of 'backsheesh' and cheroots; he helped to whip and tail-twist the slow-footed oxen; he roused up lazy *byle-*

wallahs (bullock-men) sleeping in their hovels, and assisted them in driving from the fields and in yoking to the cart refractory and kicking cattle. He stirred up with the long pole the *peons* (keepers) in charge of the road-side travellers' bungalows at which we halted, aiding these officials in chasing, slaughtering, and 'spatch-cocking' the ever-waiting-to-be-killed-and-cooked gaunt and fleshless *morghee* (fowl); he saw that the chatties for the bath were not filled with the very dirtiest of tank water; that the numerous and hard-biting insects, out and taking the air from their thickly populated homes in the crevices of cane-bottomed chair and bedstead, met with sudden and violent death; and lastly, that no man's hand but his own should be put into master's money-bag and stores.

But as all things come to an end more or less, so did Sam's career with me actually terminate. My wife and family came 'out' from England. The 'Mem Saab,' sometimes even the 'Misse Saab,' took bazaar 'count; the current bachelor rates for chillies, cocoa-nuts, first and second sorts *wrice*, gram, and such-like necessities underwent a fall. Sam's occupation and gain were gone. He quitted my homestead under this new and unprofitable régime. 'I discharge you, sar!' said he; and away he went, I know not where.

HELENA, LADY HARROGATE.

CHAPTER XI.—AN UNEXPECTED MEETING.

THE *De Vere Arms* at Pebworth, fourth-rate hotel though it necessarily was in a place where any hotel of the first or even of the second magnitude would have been as an oak in a flower-pot, was well and neatly kept. There was the commercial connection, and there was the county connection, both dear to the landlord, but on grounds wholly dissimilar. Biggles had been butler to the present, under-butler and knife-boy to the late Earl of Wolverhampton; and had he but had his own way, the *De Vere Arms* would have been strictly the family hotel which its address-cards proclaimed it, and the obnoxious word 'commercial' would have found no place there.

Mr Biggles, however, was in the position of one of those unfortunate managers of English country theatres who tell their friends, perhaps truly, that they would play nothing, save the legitimate drama, if they could help it. They cannot help it, and scared by the dismal spectre of Insolvency, they shelve Shakspeare in favour of newer idols of the public. So did Biggles and worthy Mrs B. to boot lay themselves out in practice to secure the lucrative custom of the ready-money, constantly moving, commercial gentlemen, while in theory devoting all their loyalty to those of their patrons who came in their own carriages, with armorial bearings on their panels and liveried servants on the driving-seat.

To this hostelry was borne, in Sir Gruntley Pigbury's carriage, the insensible form of Jasper Denzil, supported by the sturdy arm of Captain Prodgers, while little Dr Aulfus, on the opposite seat, kept the patient's nerveless wrist between his own thin fingers all the way from the race-course to the inn. Then Jasper, amidst spasmodic gaspings from the landlady and sympathetic exclamations from the chambermaids, was carried into the *De Vere Arms* and established in one of the best rooms, whence were summarily dislodged the effects of some well-to-do customer who had had a horse in the race, but who was unlikely under the circumstances to resent the invasion of his apartment. Jack Prodgers and the

doctor seemed to have taken joint possession of the invalid; the former as *prochain ami* (and it is to the credit of such ne'er-do-wells as Captain Prodgers that the very wildest of them never do leave a friend untended in a scrape), and the other professionally.

Other friends came not. Lord Harrogate did indeed tap at the door, and so did four or five officers of the Lancer regiment, but contented themselves with an assurance that Jasper was in no immediate danger. And when Blanche Denzil's tearful entreaties induced the Earl to solicit admittance to the sick-room for her at least, the surgeon went out and politely deprecated her entrance. Anything which might excite the patient should, he truly said, be as far as possible avoided. It was not exactly possible just yet to ascertain the amount of damage done; but he, the doctor, anticipated no serious consequences. And with this assurance the poor sister was compelled to be content. They say that every educated man of fifty is a fool or a physician. Jack Prodgers had seen the light some half-century since, and his worst enemies—the men whose cash he pouched at play—would not have taxed him with folly.

'Now, doctor,' he said quietly, 'don't you think the best we can do for the poor fellow is to get his left shoulder into the socket again before the muscles stiffen?'

The surgeon winced. He knew by the cursory examination he had made that no bones—unless it might be the collar-bone, an injury to which is not always promptly ascertained—were broken; but here, annoying circumstance! was a dislocation which he had left to be discovered by an outsider to the profession.

'Bless my soul!' he exclaimed, adjusting his spectacles, 'so it is. We have no time to lose.'

As it was, time enough had been lost to bring about a contraction of the muscles, that rendered it necessary to call in the aid of James the waiter and Joe the boots, before the hurt shoulder could be reinstated in its normal position.

The pain of the operation roused Jasper from his stupor. He moaned several times and stirred feebly to and fro, and when the wrench was over, opened his eyes and gazed with a bewildered stare about him. Very pale and ghastly

he looked, lying thus, with the blood slowly oozing from a cut on his right temple, and his hair stained and matted. They sprinkled water on his face and put brandy to his lips; but he merely groaned again, and his eyes closed.

‘That’s a very ugly knock on the temple; I hope there’s no more mischief,’ said the doctor in a whisper, but speaking more openly than medicine-men, beside a patient’s bed, often speak to the laity.

Jack Prodgers shook his head. He was a man of experience, and had in his time seen some prompt and easy recoveries, and other cases in which there was no recovery at all. It was with some remorse that he looked down at the bruised and helpless form lying on the bed. His heart had been case-hardened by the rubs of a worldly career, but there was a soft spot in it after all, and it was with sincere joy that he saw at length the sick man’s eyes open with a glance of evident recognition, while a wan smile played about his lips.

‘I say, Jack,’ said Jasper feebly, ‘we’re in a hole, old man, after all’—— Then he fainted.

‘Nothing the matter with his reason, thank goodness! It was the shock to the brain I feared the most for him,’ said the doctor, as again brandy was administered.

The regular clock-work routine of social machinery must go on in despite of accidents, and accordingly the down-train reached Pebworth at 3.40 (or, to tell the truth, a few minutes behind time) with its usual punctuality. There was no omnibus, whether from the *De Vere Arms* or from the opposition or *White Hart* hotel, in waiting at the station, wherefore the few arrivals had to consign their bales and bags and boxes of samples to the wheelbarrows of porters, for conveyance to whichever house of entertainment they designed to patronise. Amongst these was a thickset middle-aged man, with trim whiskers, a dust-coloured overcoat, a slim umbrella, and a plump black bag, which he preferred to carry as he trudged from the station to the hotel.

There was nothing very noteworthy about the new-comer, who was neatly dressed in black, and wore a hat that was just old enough to have lost its first tell-tale gloss, except that he had evidently striven to look some years younger than the parish register would have proclaimed him. Thus the purplish tint of his thick whiskers and thinned hair, heedfully brushed and

parted so as to make the most of it, savoured of art rather than nature. His cravat too, instead of being black, was what haberdashers call a scarf of blue silk, of a dark shade certainly, but still blue, and was secured by a massive golden horse-shoe. Glittering trinkets rattled at his watch-chain, and his boots were tighter and brighter than the boots of men of business usually are. There is or ought to be a sort of fitness between clothes and their wearer, but in the case of this traveller, obviously bound for the *De Vere Arms*, no such fitness existed. That cold gray eye, those deeply marked crow's-feet, the coarse mouth, and mottled complexion, consorted ill with the pretensions to dandyism indicated by a portion of their owner's attire. Altogether, the man might have been set down as a corn-doctor, a quack, a projector of bubble companies, or possibly an auctioneer whose hammer seldom fell to a purely legitimate bid in a fair market.

As the stranger drew near to the hotel, having inquired his way once or twice from such of the natives as the great attraction of the day had not allured to the race-course, a carriage dashed past him at a very fast pace indeed, and drew up with a jerk in front of the *De Vere Arms*. The gentleman who alighted from it, tall, and of a goodly presence, lingered for an instant in the doorway to give some order to his servants. As he did so, his eyes encountered those of the traveller freshly arrived by the train, and who by this time was beneath the pillars of the porch. Sir Sykes Denzil, for it was he whose carriage had just brought him in hot haste to the place where his son lay ill, started perceptibly and hesitated, then turned abruptly on his heel and disappeared within the hotel, greeted by the obsequious Mr and Mrs Biggles.

Recognition, as we can all avouch, is in the immense majority of cases simultaneous, one memory seeming as it were to take fire at the spark of recollection kindled in the other. In this instance such was not exactly what occurred. Yet the traveller with the bag was perfectly certain that he had seen before the tall gentleman who had started at the sight of him, and that a diligent searching of the mental archives would elicit the answer to the riddle.

'Have I written or telegraphed to order rooms here?' repeated the new arrival testily, after the flippant waiter who came, flourishing his napkin, to see what the stranger wanted. 'No, I have not. And to judge by the size of your town, my friend, and the general look of affairs, I should say that on

any other day of the year but this such a precaution would be wholly superfluous.'

The waiter, who had been slightly puffed up by the ephemeral vogue of Pebworth and its chief hotel, took the rebuke meekly. 'Would you step into the coffee-room, sir?' he said. 'I'll ask Mrs Biggles about accommodation likely to be vacant. Any name I could mention, sir?'

'Name—yes, Wilkins,' returned the traveller, pushing open the door of the coffee-room, in which, at various tables, some dozen of sporting-men were making a scrambling meal. One or two of these looking up from their plates, nodded a greeting, with a 'How d' ye do, Wilkins?' or 'How goes it, old fellow?' salutations which the recipient of them returned in kind. Then the waiter bustled in to say, more respectfully than before, that so soon as No. 28 should be vacated by a gentleman leaving by the 6.25 train, it would be at the disposal of Mr Wilkins. Further, here was a note for Mr Wilkins; into whose hand he proceeded to thrust a half-sheet of letter-paper, roughly folded in four, and containing but some two or three lines of blotted handwriting. 'If you will so far oblige me'—thus ran the words, shaky and blurred as to their caligraphy, but tolerably legible—'I shall be glad of a few moments' interview with you, at once if not inconvenient, in No. 11. I will not detain you.'

There was no signature, but no reasonable doubt could exist in the mind of Mr Wilkins as to the note having been penned by the owner of the carriage that had so lately driven up to the door of the *De Vere Arms*.

'Why, this is taking the bull by the horns,' said Mr Wilkins, as he rose to obey the summons.

CHAPTER XII.—IN NO. XI.

No. 11 was a sitting-room of a class peculiar to those old-fashioned inns which are rapidly being improved off the length and breadth of Britain, large, low-ceiled, with a sloping floor that attained its highest elevation beside the broad bay-window. A dark room, it must be confessed, and an airless, but snug and warm on winter-nights, when the glow of the firelight combined with the lustre of many wax-candles to defy the storm and blackness without. There had been jovial dinners in that room, and drawing

together of arm-chairs around the huge fireplace, and tapping of dusty magnums of rare old port, and calling for more punch as the night waned, in those hard-living days of which so many of us innocent, pay the penalty in neuralgia and dyspepsia.

In No. 11 stood Sir Sykes, pale but resolute. The traveller with the black bag came in, and for the second time their eyes met. 'You wished to see me, sir,' began Mr Wilkins, with a slight bow. 'Ah! I remember you now, sir, as it happens,' he added in a different tone; 'remember you very distinctly indeed, Mr'——

'Hush!' interrupted Sir Sykes, with uplifted fore-finger. 'A place like this is the very last in which to mention anything best left unspoken—the very walls, I believe, have ears to hear and tongues to tattle. I am Sir Sykes Denzil, of Carbery Chase, within a very few miles of this, at your service, Mr Wilkins.'

'Sir Sykes Denzil! Well, *this is* a surprise,' exclaimed the owner of the name of Wilkins wonderingly, and yet with a sort of dry humour mingling with his evidently genuine astonishment. 'Dear me, dear me! They say the world is very little, and people constantly meeting and jostling in it; but I never so thoroughly realised the truth of the saying as I do now. So I've the honour of talking to Sir Sykes Denzil, when I thought I was addressing'——

'Be cautious, sir,' interposed the baronet, with an energy that impressed the other in spite of himself. 'Let us have no reference, if you please, to a past that is dead and buried. I sent for you, certain as I was that sooner or later your memory must recall me to your remembrance, and well aware too how easily you could learn who I was here.'

'No great trouble about that, Mr—I mean Sir Sykes,' rejoined the traveller smirkingly. 'The people seem to know you well enough, and any fellow in the stable-yard would have told me whose was the carriage with the brown liveries.'

'And having met and recognised one another,' said Sir Sykes, 'on what footing is our future intercourse to be conducted? We are not as we once were, lawyer and client, and'——

'No, Sir Sykes, I grant you that; but we might be,' returned Mr Wilkins, rubbing his fleshy hands together, as though they had been two millstones

between which the bones of suitors might be ground to make his bread. ‘You can’t, a man of your landed property—I’ve heard something as to your acreage, and could give a shrewd guess as to your rent-roll—be without law business. Devonshire isn’t Arcadia, I suppose. Are there not leases to draw, inclosure bills to promote, poachers to prosecute, paths to stop up, bills to file, actions to bring, defend, compromise? Ten to one, some of your best farms are let on leases of lives, and—— But no matter! You’ve your own legal advisers; hey, Sir Sykes?’

The baronet bowed coldly by way of assent.

‘Pounce and Pontifex, of Lincoln’s Inn—I know,’ pursued the unabashed lawyer. ‘A brace of respectable twaddling old stagers. There was a saying, soon after I got my articles, as to that firm, to the effect that Pounce and Pontifex were fit for a marriage settlement, a will, and a Chancery suit, and that was about all. If you care about raising your rents, crushing an enemy, or gratifying a whim—and most rich men have a hankering after one or other of these fancies—why, you’ll need a brisker counsellor at your elbow than the jog-trots of Lincoln’s Inn.’

Again the baronet bent his head, and his eyes moved towards the door. Mr Wilkins noted their movement.

‘You hardly derived a fair judgment of my capabilities,’ he said, ‘by the little I had to do in that Sandston business’——

‘Again I ask you, sir, to make no mention of that subject. It—it is naturally painful to me—and—and’—— Sir Sykes here fairly broke down.

The lawyer’s eyes twinkled as he saw his advantage. ‘So long as *you* remember it, Sir Sykes,’ he made haste to say, ‘I shall be only too happy to forget the whole concern. What was that story about the organ-blower and Handel? “Shan’t it be ‘we,’ then?” said the fellow, when the great organist couldn’t get a note out of his instrument for want of the necessary but humble bellows. And the musician was compelled to acknowledge that there was a sort of partnership between the man who fingered the stops and the man who raised the wind. I’m in no hurry. Think it over. I have a client to see here to-day; but perhaps you will let me have a word with you before you drive back to Carbery Chase.’

A long deep line, which might have been mistaken for the furrow of some old sword-cut, running from the angle of the mouth obliquely upwards, became visible in the baronet's comely face as he listened. He was one of those men who can better endure misfortune than disrespect, and to whom the bitterest sting of ruin is the withdrawal of the deference and lip-service which environ them. But it was in an amicable tone that he made answer: 'I shall be happy to pursue our conversation, Mr Wilkins, to-day or at any time which you may deem suitable. At present, however, you will excuse me if I leave you. My son, Captain Denzil, has been hurt—badly hurt, I fear, in the steeplechase to-day, and I have been called here to see him, where he lies, in this very hotel.' And the baronet moved towards the door.

'Hurt, is he?' exclaimed Mr Wilkins, with inconsiderate roughness. 'Ah, then, I shall look to you, Sir Sykes, to indemnify me in case'——

Then came an awkward pause. The solicitor was a remarkably plain-spoken man, but he did not quite like to say, 'in case your son's accident prove fatal,' and so stopped, and left his eloquent silence to complete his words. Sir Sykes, with his hand on the door, turned, astonished, upon the attorney.

'What, pray, have you to do with the illness or the recovery of Captain Denzil?' he asked in evident ill-humour. He had borne up to this with Mr Wilkins, but the lawyer's interference with regard to his son appeared to him in the light of a gratuitous piece of insolence.

'Simply,' returned Mr Wilkins, thrusting his hand into an inner pocket of his coat, 'because I am the holder of certain acceptances, renewed, renewed afresh, and finally dishonoured; acceptances amounting, with expenses, to a gross amount of—shall we say some eleven or twelve thousand, Sir Sykes? Nearer the twelve than the eleven, I suspect. A flea-bite of course to a gentleman of your fortune, but a very important sum to a plain man like yours truly.'

'I have been put to heavy expense, very heavy, for my son's debts,' said Sir Sykes, almost piteously. 'I have paid every'——

'Now, my very good sir,' interrupted the attorney, 'don't, I beg you, don't fall into the common error of fathers, and imagine that your own particular son is either a miracle of ingenuous candour or a prodigal worse than his neighbours. You think that you've paid all his liabilities, Sir Sykes, and no

doubt you have paid all you knew of. But as a man of the world, if not as a parent, you ought to be aware that nobody ever did tell all that he owed—excess of modesty, perhaps! They always leave a margin, these interesting penitents; and in this case, as you will see by these documents’ (and Mr Wilkins produced several pieces of stamped paper), ‘the margin is tolerably ample.’

The baronet was now thoroughly roused to wrath. He strode to and fro with frowning brow and hands that were fast clenched together, then walked to the window and stood still, idly tapping the panes with one white finger, on which there glistened a great diamond that had been an heirloom at Carbery Chase before ever a Denzil crossed its threshold.

‘I’ll not give him a shilling or leave him a shilling!’ he said in a voice that quivered with anger. ‘Carbery Chase is my very own, and I can deal with it as I please. My daughters at anyrate have deserved better of me than that thankless graceless boy.’

Sir Sykes, under the influence of this new emotion, seemed to have forgotten the lawyer’s presence, or merely to regard Mr Wilkins in the light of the impartial Chorus in a Greek tragedy; but the attorney, who was by no means pleased by the turn which the affair seemed to be taking, intervened.

‘Come, come, Sir Sykes. It’s natural that you should be annoyed at having such a heavy bill presented, when you thought it settled. But between ourselves, boys will be boys. The captain has turned over a new leaf, and rely on it he will be a credit to you yet. I’ve a pretty wide acquaintance amongst wild young gentlemen of his kind, and I give you my word I don’t know one who is more wide-awake. He had paid his ’prentice fees, and that smartly; but I expect before I die to hear of him as an ornament to the bench of magistrates and perhaps a county member. As for these bills and notes of hand’——

‘I’m not liable for a sixpence!’ exclaimed Sir Sykes petulantly. ‘My son may go through the Court if he chooses, and perhaps will learn a wholesome lesson from the exposure, which’——

‘Fie, fie, Sir Sykes!’ broke in the lawyer. ‘A coat of whitewash, believe me, sticks to a youngster’s back to that extent that no amount of scrubbing can get rid of it. Fume and fret as you please, you know, and I know, that you

mean Captain Jasper to have Carbery after you, and to keep the place in the Denzil line. Better so, than to have so fine an estate sold or cut in two for division between your daughters' husbands. And the captain won't bear the 'bloody hand' in his escutcheon the better because he has been made an insolvent in his youth. As for these claims, I don't press for an immediate settlement; not I; I don't exact my pound of flesh down on the nail, Sir Sykes.'

There was a hard struggle in the baronet's breast. Time had been given him for reflection, and he had used it. To hear of his son's extravagance, of his son's deceit, and from such lips, was bad enough. To be compelled to endure the familiarity of the lawyer's manner was to have to swallow a still more bitter pill. He could remember Mr Wilkins of old, blunt and jocose certainly, but by no means so jaunty in his bearing as he now was, although Sir Sykes had not then been the rich county magnate he had blossomed. He felt, and writhed as he felt, that it was the attorney's sense of his hold upon him by reason of his knowledge of his past life, which had emboldened Mr Wilkins to deal with him as he had done. But the most provoking feature of the affair was that Sir Sykes felt that this man's advice, coarsely and offensively administered as it was, yet contained a solid kernel of truth. Jasper was by no means a model son. He had committed fearful follies, and incurred debts which even the Master of Carbery had thought twice before discharging. His profligacy was redeemed by no brilliant talents, softened by no affectionate qualities. There are spendthrifts who remain lovable to the last, as there are others who dazzle the world by the glitter of their wit or valour. To neither category did the graceless offspring of Sir Sykes belong. And yet, in spite of his occasional menaces on the subject of his will, the baronet felt that national manners and family pride combined to constitute a sort of moral entail, of which Jasper was to reap the benefit.

'I must see my son,' said Sir Sykes smoothly, after a pause; 'and when I have time to think over the matter, Mr Wilkins, I will write to you appointing as early an interview as possible. In the meantime I feel assured that you will see the propriety of not urging personally your claims on Captain Denzil in his present condition.'

Mr Wilkins was amenity itself. He would but eat a morsel in the coffee-room, he said, and would then go back to London by the next train,

confident that he could not leave his interests in better hands than those of Sir Sykes.

‘The old address, sir! You used to know it well enough!’ said the lawyer with a leer, as he took the hand which the baronet did not dare to refuse in sign of friendship; and so they parted.

COAL AND ITS PRODUCTS.

IN an article which appeared in this *Journal* in August 1876, entitled *The Age of the World*, we endeavoured to explain how coal was produced, and how it might be regarded simply as stored-up heat and light, derived from the sun ages ago.

Apart from the varied uses of coal in its ordinary state, we owe an immense deal to the products which by chemical means we obtain from it; and it is our purpose in this paper to briefly review these products, and to shew how we have adapted them to our several wants.

The manufacture of gas is undoubtedly the most important feature in the modern history of coal. Natural reservoirs of inflammable air exist in many parts of the world, and have in many cases been turned to profitable account. In China, for instance, the evaporation of salt has for many years been carried on by the heat obtained by the combustion of gas which issues from the ground. Streets and buildings there have also been lighted by the same means. In our own country too, such eruptions of natural gas—which have generally manifested themselves during the operation of well-boring—have not been uncommon. But the gas so obtained is not the same as that which we get from the distillation of coal, although it forms one of its chief constituents. It is commonly called marsh-gas, from its constant presence in bogs and places where decaying vegetable matter abounds. The treacherous Will o' the Wisp owes its origin to this gas. It also issues in large quantities from coal-beds, and diluted with air forms the dreadful compound called 'fire-damp.'

The first recorded experiment relating to the production of true coal-gas was as early as the year 1660, when a country clergyman distilled some coal, collected the gas in bladders, and burnt it from a jet, for the amusement of his friends. Although this very suggestive experiment was communicated to the Royal Society, no action seems to have been taken upon it until the beginning of the present century, when the matter seems to have attained a more practical form. At this time one or two factories in Manchester and Birmingham were for the first time lighted with gas. The

idea of illuminating an entire town by means of a chemical vapour seems to have met with much ridicule, and it was found necessary to employ lecturers to go about the country to shew people how such an apparent impossibility could be carried out. However, in spite of much opposition, part of London was lighted by gas in 1812; and three years later, Paris adopted the same system. The delay in the acceptance of gas-making among the industrial arts was no doubt largely due to the expressed opinion of several eminent chemists and others, who considered that such a mode of lighting our towns could never be realised, because of the supposed danger which it involved. Modern experience teaches us that it is at once the cheapest as well as the safest mode of illumination that we can as yet command. In the manufacture of gas, the coal is placed in iron retorts, which are subjected to a high temperature for about six hours, when the operation is finished, and the retorts are ready for a fresh charge. A residue of nearly pure carbon, in the form of coke, remains in the retort, whilst the varied products of the distillation are carried off by pipes into suitable receptacles. For the sake of convenience, we will at present name only three of these products—ammoniacal liquor, tar, and the gas itself. The first is the principal source of ammonia, one of the most useful substances known. It may be almost said of ammonia, as it has been remarked of sulphuric acid, that the prosperity of a country may be known by the quantity which it consumes. It is used by colour-makers, calico-printers, and in the manufacture of most of the textile fabrics; in cleansing and extracting grease from various kinds of cloth, in the preparation of leather, in galvanising iron, and in pewtering. The chemist would be almost helpless without its aid; whilst in medicine it is used in about twenty different forms as a most valuable stimulant. It is almost needless to say that ammonia was in general use long before the era of gas-manufacture, for life could hardly go on without it. In fact its very name is derived from its manufacture hundreds of years ago from animal refuse in a district of Libya where the deity Jupiter Ammon was worshipped. The old alchemists too obtained it from the distillation of deer's horns; hence one preparation of it is still called spirit of hartshorn. There are many other sources of ammonia, for its presence in nature is universal; but all have sunk into insignificance since the gas-works have yielded such plentiful supplies.

Coal-tar in its crude state is not of very great importance, its use being confined to such rough work as the water-proofing of boats and the painting

of outhouses and the like. But in the hands of the chemist its applications cannot be lightly regarded, in fact its distillation is of sufficient importance to form a distinct branch of trade. In this process coal-tar is separated into three different products—naphtha (which in a rectified state is the benzol of commerce); heavy or creosote oil, which is used almost exclusively for the preservation of railway sleepers; and the residue pitch. The last is of great use to shipbuilders, and has more recently found employment in the preparation of asphalt roofing and paving. But naphtha is by far the most important of the three substances, if it were only for its use as a solvent for both india-rubber and gutta-percha. No doubt, failing this, other solvents for caoutchouc would have been found; but naphtha is a particularly cheap and effective menstruum for the purpose; and when we consider the varied uses to which india-rubber and gutta-percha are now applied—from elastic hosiery to submarine cables—we must acknowledge that naphtha is a valuable addition to our manufacturing resources. It is a significant circumstance that the date of the introduction of manufactured india-rubber (by Mr Mackintosh) follows the general adoption of gas-lighting by only a few years. Previous to this, india-rubber was imported merely as a curiosity, its first use being to obliterate pencil-marks, for which purpose it was once advertised in London at the modest price of six shillings per square inch.

Besides its use as a solvent, benzol is of particular importance in yielding, when treated with nitric acid, a substance called aniline. The discovery of aniline is one of the most remarkable triumphs of chemistry, as applied to the advancement of a manufacturing industry. (Before the date of coal-tar it was obtained from indigo, and the name it bears is the Portuguese for that colour.) The production of aniline caused quite a revolution in the various trades which are dependent in any way upon the colour-manufacturer; for lithographers, paper-stainers, calico-printers, and especially dyers, owe their most brilliant tints to its aid. The various dyes which are now commonly retailed for household use are also derived from the same source. Aniline is an almost colourless liquid, of a peculiar vinous odour, which after exposure to the air, changes to a dark resinous matter. The treatment which it undergoes in producing the various colours (and nearly every colour of the rainbow can now be obtained from it), is of too complicated a nature to be of any interest to the general reader. Magenta, the advent of which some years back many of our readers will remember, was the first aniline dye which appeared. The other colours have followed in quick succession,

nearly all of them being the subjects of one or more patents. It is questionable whether all these colours are strictly permanent; but it is a pleasing thought that the hues which in one form or another existed at a period long before mankind had a place in nature, are now reproduced for man's delight and benefit.

Another very important product of gas-tar is carbolic acid, which is also largely employed for dyeing purposes. Its value as a disinfectant is too well known to need recapitulation here; but we may mention that its use as a preventive of disease was most abundantly proved during the last epidemic among our cattle. It is in general use in our hospitals, not only as a disinfectant, but also as an antiseptic both in the dressing of wounds and in the treatment of various skin diseases. Carbolic acid also yields a substance called picric acid, which, on account of its explosive properties when combined with potassium, has been proposed as a substitute for gunpowder. There are many other substances derived from the distillation of coal-tar, but at present they are only of interest to the experimental chemist.

A ton of coals will produce a chaldron of coke, twelve gallons of tar, ten gallons of ammoniacal liquor, and nearly ten thousand feet of gas. A consideration of these figures, with a due regard to what we have said as to the value of the various chemical products obtained by distillation, will enable our readers to understand why gas companies can shew such good balance-sheets. Much has been written as to the possible exhaustion, after one or two centuries, of the British coal-fields. This is a question upon which it is next to impossible to form any reliable opinion. Should the coal-supply actually fail, it is more than probable that as science is extended, a new source of light and heat may be developed. A cheap and ready means of producing electricity, as we have in a former article endeavoured to shew, would at once solve the problem, and it is within the bounds of reason that to this agency the future races of the earth will look for the two most common necessities of existence.

MALAPROPOS.

CHARLES DICKENS once wrote to a friend: 'I have distinguished myself in two respects lately. I took a young lady unknown down to dinner, and talked to her about the Bishop of Durham's nepotism in the matter of Mr Cheese. I found she was Mrs Cheese. And I expatiated to the member for Marylebone, Lord Fermoy—generally conceiving him to be an Irish member—on the contemptible character of the Marylebone constituency and Marylebone representatives.' Two such mishaps in one evening were enough to reduce the most brilliant talker to the condition of the three 'insides' of the London-bound coach, who beguiled the tedium of the journey from Southampton by discussing the demerits of William Cobbett, until one of the party went so far as to assert that the object of their denunciations was a domestic tyrant, given to beating his wife; when, much to his dismay, the solitary lady passenger, who had hitherto sat a silent listener, remarked: 'Pardon me, sir; a kinder husband and father never breathed; and I ought to know, for I am William Cobbett's wife!'

Mr Giles of Virginia and Judge Duval of Maryland, members of Congress during Washington's administration, boarded at the house of a Mrs Gibbon, whose daughters were well on in years, and remarkable for talkativeness. When Jefferson became President, Duval was Comptroller of the Treasury, and Giles a senator. Meeting one day in Washington, they fell to chatting over old times, and the senator asked the Comptroller if he knew what had become of 'that cackling old maid, Jenny Gibbon.' 'She is Mrs Duval, sir,' was the unexpected reply. Giles did not attempt to mend matters, as a certain Mr Tuberville unwisely did. This unhappy blunderer resembled the Irish gentleman who complained that he could not open his mouth without putting his foot in it. Happening to observe to a fellow-guest at Dunraven Castle, that the lady who had sat at his right hand at dinner was the ugliest woman he had ever beheld; the person addressed expressed his regret that he should think his wife so ill-looking. 'I have made a mistake,' said the horrified Tuberville; 'I meant the lady who sat on my left.' 'Well, sir, she is my sister,' was the response to the well-intentioned fib; bringing from the desperate connoisseur of beauty the frank avowal: 'It can't be helped, sir,

then; for if what you say be true, I confess I never saw such an ugly family in the course of my life!’

An honest expression of opinion perhaps not so easily forgiven by the individual concerned, as that wrung from Mark Twain, who, standing right before a young lady in a Parisian public garden, cried out to his friend: ‘Dan, just look at this girl; how beautiful she is!’ to be rebuked by ‘this girl’ saying in excellent English: ‘I thank you more for the evident sincerity of the compliment, sir, than for the extraordinary publicity you have given it!’ Mark took a walk, but did not feel just comfortable for some time afterward.

One of the humorist’s countrymen made a much more serious blunder. He was a married man. Going into the kitchen one day, a pair of soft hands were thrown over his eyes, a kiss was imprinted on his cheek. He returned the salute with interest, and as he gently disengaged the hands of his fair assailant, asked: ‘Mary, darling, where is the mistress?’ and found his answer in an indignant wife’s face. ‘Mary darling’ had gone out for the day, and the lady of the house intended by her affectionate greeting to give her lord a pleasant surprise. He got his surprise; whether he thought it a pleasant one he never divulged, but that kitchen knew Mary no more.

A stout hearty-looking gentleman one day made his way from the dock-side at Plymouth to the deck of a man-of-war newly arrived from abroad, and desired to be shewn over the ship. Most of the officers were on shore, and the duty of playing cicerone devolved upon a young midshipman. He made the most of his opportunity, and to have a lark at the expense of the elderly gentleman as he shewed him round, he told him how the capstan was used to grind the ship’s coffee, the eighteen-ton guns for cooling the officers’ champagne, the main-yards for drying the Admiral’s Sunday shirts, and many other things not generally known. When the gentleman had seen all he wanted to see, he handed a card to his kind instructor, saying: ‘Young gentleman, you are a very smart youth indeed, and full of very curious information; and I trust that you will see there is no mistake in this card of mine finding its way to your captain.’ The middy glanced at the bit of pasteboard and read thereon the name ‘Ward Hunt;’ but before he could thoroughly realise the situation, the First Lord of the Admiralty, with a parting nod and pleasant smile, had gone.

Another story, illustrating the awkward results that come of letting the tongue wag freely under a misapprehension regarding other folk's identity, is told of a London tailor. An aristocratic customer noted for dressing in anything but aristocratic fashion, called to pay his bill. The tailor's new manager, after receipting the account, handed it back with a sovereign, saying: 'There's a sovereign for yourself, and it's your own fault it isn't two. You don't wear out your master's clothes half quick enough. He ought to have had double the amount in the time; it would be worth your while to use a harder brush.'

'Well, I don't know,' said his lordship, smiling; 'I think my brush is a pretty hard one too; his lordship complains of it anyhow.'

'Pooh! Hard! Not a bit of it! Now I'll put you up to a dodge that'll put many a pound in your pocket. You see this piece of wood—now that's roughened on purpose. You take that, and give your master's coat a good scrubbing with it about the elbows and shoulders every day; and give the trousers a touch about the knees, and it will be a good five pounds a year in your pocket. We shan't forget you.'

'You are very kind,' quoth the enlightened gentleman. 'I will impart your instructions to my valet, though I fear while he remains in my service he will not be able to profit by them, as I shall not trouble you with my custom. I wish you good-day.'

We read in Lord Eldon's Journal: 'The most awkward thing that ever occurred to me was this. Immediately after I was married I was appointed Deputy Professor of Law at Oxford, and the Law Professor sent me the first lecture, which I had to read immediately to the students, and which I began without knowing a single word that was in it. It was upon the statute applying to young men running away with maidens. Fancy me reading with about one hundred and fifty boys and young men all giggling at the Professor! Such a tittering audience no one ever had.' The comical coincidence may have been an accidental one; but as the Law Professor must, like the students, have known that his deputy ran away with his Bessie, the chances are against it. The great lawyer was fated to be reminded of the romantic episode of his life. A client whose daughter had been stolen from him, insisted upon the jury being told that a man who could run away with another man's daughter was a rascal and a villain, and

deserved to be hanged. 'I cannot say that,' said Scott. 'And why not, Lawyer Scott—why not?' inquired the irate father. 'Because I did it myself!' was the unanswerable reply.

After doing his office for a young couple, a clergyman was inveigled into proposing the health of bride and bridegroom at the wedding breakfast. He wound up a neat little speech by expressing the hope that the result of the union of the happy pair might prove strictly analogous to that of the bride's honoured parents. The groom looked angry, the bride went into hysterics, the bridesmaids blushed and became interested in the pattern of the carpet, the master of the house blew his nose with extraordinary violence, and the speaker sat down wondering at the effect he had created; till his better-informed neighbour whispered that the lady was not the daughter of the host and hostess, but a niece who came to live with them when her mother and father were divorced.

During Mr Gladstone's Premiership, Sir George Pollock called one morning in Downing Street to thank the Prime-minister for making him governor of the Tower. A cabinet council had just assembled; but rather than keep the veteran waiting, Mr Gladstone invited him into the council-chamber and introduced him to his colleagues. Sir George entertained his new acquaintances with a tedious story about a nobleman who had been detected cheating at cards, ending his narration with: 'They turned him out of all the clubs he belonged to; even the Reform would have nothing more to say to him!' A way of proving the enormity of the card-player's offence that must have pleased his hearers amazingly, since all or nearly all of them were members of that famous Liberal club.

The old governor sincerely meant what his words implied. Such is not always the case with utterers of malapropos things. When a note was handed to Dr Fletcher in his pulpit intimating that the presence of a medical gentleman, supposed to be in the church, was urgently required elsewhere, the preacher read the letter out, and as the doctor was making for the door, fervently ejaculated: 'May the Lord have mercy on his patient!' A Scotch minister exchanging pulpits with a friend one Sunday, was accosted after service by an old woman anxious to know what had become of her 'ain minister.' 'Oh,' said he, 'he is with my people to-day.' 'Indeed, indeed,' said the dame; 'they'll be getting a treat the day!' As flattering a remark as that of the wife of a popular lecturer, who on her lord telling her he was going to

lecture at Sheffield, exclaimed: 'I'm so glad; I always hated those Sheffield people.'

Epitaph writers sometimes display a talent for this kind of *double-entendre*. A couple of specimens will suffice. The first from Arbroath, running: 'Here lie the bodies of John, William, Robert, and David Matthews, who all died in the hope of a glorious resurrection—excepting David.' The other from an American burying-ground:

Here lies the mother of children five;
Two are dead and three are alive;
The two that are dead preferring rather
To die with their mother than live with their father.

Although a high authority insists that the lunatic and the lover are of imagination all compact, it would not enter an ordinary lover's head to tell his mistress that loving her was synonymous with madness, as Steele did when he wrote to his dear lovely Prue: 'It is the hardest thing in the world to be in love and yet attend business. As for me, all who speak to me found me out, and I must lock myself up, or other people will do it for me;' but fair Mistress Scurlock doubtless took the dubious flattery in as good part as the great animal painter took the king of Portugal's odd greeting: 'Ah, Sir Edwin, I am glad to see you; I'm so fond of beasts.' An unpleasant way of putting the thing was innocently adopted by the New York car-driver, who, blissfully ignorant that his interlocutor was Mr Beecher, replied to that gentleman's query whether he did not think it possible to dispense with running the cars all day on Sunday: 'Yes, sir, I do; but there's no hope for it so long as they keep that Beecher theatre open in Brooklyn; the cars have to run to accommodate that.'

An American newspaper says: 'The enthusiastic choir-master who adopted *Hold the Fort* as a processional hymn, has been dismissed by the minister, who considered it personal when the choir burst forth:

See the mighty host advancing,
Satan leading on!

A similar objection might have been raised to the Maine county commissioners quoting Watts's lines:

Ye sinners round, come view the ground
Where you will shortly lie,

when inviting certain lawyers to inspect the new court-house; although they had less reason to complain than Lord Kenyon and Justice Rooke, who while on circuit, came one Sunday to a little village just as the good folks were going to church; an example the two judges followed. Anxious to shew his appreciation of the unexpected honour, the parish clerk searched for a suitable psalm to sing before the service; and at the proper time gave out the first two verses of the fifty-eighth psalm, and the congregation sang:

Speak, O ye judges of the earth,
If just your sentence be;
Or must not innocence appeal
To heaven from your decree.
Your wicked hearts and judgments are
Alike by malice swayed;
Your griping hands, by mighty bribes,
To violence betrayed.

Here the congregation awoke to the meaning of what they were singing, and left the clerk and the children to offend the ears of the legal dignitaries with:

To virtue, strangers from the womb,
Their infant steps went wrong;
They prattled slander, and in lies
Employed their lisping tongue.
No serpent of parched Afric's breed
Does ranker poison bear;
The drowsy adder will as soon
Unlock his sullen ear.

The performance unlocked the tongues of the astonished judges at any rate; and the churchwardens had some difficulty in convincing them that the apparent insult arose out of the stupidity of the well-meaning clerk.

THEODOR MINTROP.

‘THE poor world is almost six thousand years old, and in all this time there was not any man died in his own person, namely, in a love-cause.’

I cannot help recalling Rosalind’s words as I look at the photograph before me; the history of its original so completely disproves her saucy speech. In my hand I hold the likeness of a man of forty or thereabouts, with a noble square forehead arching above deep thoughtful eyes, a large beardless face surrounded by a heavy growth of long hair, and a thickset form denoting great personal strength. A superficial observer might call the homely portrait commonplace, and turn to gaze on the more aristocratic faces of his fellow-artists in the photographic album; but a careful scrutiny of the coarse irregular features and the broad brow impresses one with the feeling that this was no ordinary man; that a spirit dwelt within these steady eyes purer and mightier than usually falls to the lot of mortal man. But the closest inspection would still leave much untold. The indomitable energy, the heaven-sent genius, may be traced in his strong features and deep eyes; but the exquisite sensibility, the single-heartedness, the uncomplaining patience, would never be guessed.

But a short time has elapsed since he was one of us, and his story is still ringing in the hearts of his countrymen—a story so pathetic in its poverty and its triumph, so touching in its untimely close.

Theodor Mintrop, the original of the photograph, was born near the village of Werden in Westphalia. From his childhood he had an uncontrollable desire to draw, which brought nothing but censure from his elders, substantial *bauers* and petty farmers, who considered drawing an unpardonable waste of time. But the talent was not to be crushed out. In spite of opposition and discouragement, in spite of his daily hard work on his father’s farm, he practised his art whenever he had an opportunity; at first sketching rough outlines on whitewashed walls, and when he could afford it, buying pencils and paper. In time his fame as an artist spread among the simple peasantry, and even beyond his own limited circle. ‘The country Raphael,’ he was popularly called; and made a little money

occasionally by painting signs for country inns, and pictures of the Virgin and Child for the Catholics. All this time he wrought in the fields at a labourer's usual avocations; and it was a hard horny hand that in his leisure moments wielded the pencil with such surprising genius. He was waiting—waiting patiently till the tide would turn—waiting till the time would come when he could study his art and devote himself wholly to it. And thus he might have spent his entire life, his genius, like an imprisoned bird, hemmed in by sordid cares and toils, if one of these strange coincidences that so often bring the unexpected, had not occurred.

A celebrated artist, seeing some of Mintrop's drawings, was so struck by their merit, that he immediately set out for Werden, found Mintrop at the plough, and carried him back to his house in Düsseldorf, offering him every facility for studying thoroughly his beloved art.

The opportunity had come; but how long the country Raphael had waited for it! Thirty years had he repressed his ambition, and performed the duties of farm-labourer for his father and brother. No wonder a sad weariness can be traced on his features. In Düsseldorf, Mintrop went through the regular course of instruction, beginning at the very lowest class, where he, a man of thirty, sat on the same bench with young lads; but his great genius and intense application soon carried him through the class-rooms. His art had an amount of originality and freshness that seemed to breathe of his free country life at Werden. From his boyhood a great lover of fairy tales, there was a strain of grotesqueness in his works. His father, a man of an original turn of mind, had fostered his passion for the weird homely legends of the German peasantry; and to Theodor, in his imaginative youth, *kobolds* had peeped out of the earth, *nixies* had sung in the rivers. The fame of the country Raphael soon spread in Düsseldorf; art critics acknowledged his wonderful genius, and vied with one another in pointing out the grand simplicity and admirable power of his compositions. How did the untrained peasant, fresh from his rural life, bear all this homage? Simply and meekly. With reverence he regarded the wonderful new life around him, so much more polished, so much pleasanter than his old one; but the dignity of his art and his own self-respect saved him from being overborne by it. But no one guessed that under his homely and somewhat uncouth exterior such an appreciation for all that was fair and good in life existed, as the sequel of his life proved.

Behold him now at perhaps the zenith of his career; having attained the object of his desires, an artistic education; having in a few short years established a fame that many academical pupils of many years' standing had failed to win; surrounded by many friends, living in the home-circle of his first patron and dearest friend in that pleasant city on the Rhine. His future lay fair and unclouded before him, leading him on from triumph to still greater triumph. But inscrutable are the ways of Providence; God's ways are not man's ways; and the tree that promised such glorious fruit was never to reach maturity.

To the house of Geselschap (the name of the artist who had befriended Mintrop, and in whose house he lived) came one fine summer a young lady-friend. In the free unrestrained home society, Mintrop had much opportunity of becoming intimately acquainted with this young girl. He had been learning much of life as well as art since he came to Düsseldorf; but women in a higher rank than the peasants he had for thirty years been familiar with, were ever an object of peculiar interest and intense admiration to him; and the grace and amiability of this stranger soon made a powerful impression on him. For a whole long happy summer this fair young creature lived under the same roof with him, and treated the grave shy man with the playfulness and friendliness of a sister, wholly unaware of the passion she had unwittingly kindled. In short he, the hard-working country Raphael, engrossed in his art, which he pursued for itself, not for money (about which he was one of the most careless of mortals)—he, the rough Westphalian peasant, with hard hands and uncouth figure, had learned to love this gentle maiden, with all the strength of his noble patient heart.

That long happy summer passed, and the young lady returned to her friends. Shortly after, the announcement of her engagement to be married reached Düsseldorf, piercing the true heart that loved her so well. To commemorate her marriage, Mintrop composed a wonderful series of pictures, that will always link her name to his.

The 'Love of King Heinzelmännchen' they were called; seventy scenes in all, in which he, in the guise of King Heinzelmännchen, following his beloved Johanna through every incident in daily life, protects and helps her as he would fain have done in reality. True to the traditions of his youth, numbers of quaint dwarfs with long beards, pointed caps, and trunk-hose, attend on the

commands of their king; who is himself a strange weird vision with a wizened face, pointed cap, and magic wand, tipped by a burning eye. In a burgher household, these droll figures sweep and wash, bake and brew, throwing themselves into many strange contortions, in the service of Anna; the king ever with them, looking sadder and sadder; for as time goes on, a stranger from America falls in love with Johanna and carries her away across the sea. The poor gnome-king loves in vain; and when the day comes that Johanna and her lover sail away, he and his dwarfs stand sadly on the shore (for they may not cross the sea) watching the vessel till it fades from sight.

The fantastic legend is imbued with a strange humanity; and the ugly figure of the gnome-king touches our inmost sensibility with a thrill of pathos. Such was the love of Mintrop—intense, undying, and hopeless! Some things are almost too sad to bear speaking of, and the waste of affection that goes on in this world is one of them. Doubtless there were many girls in Düsseldorf equal to Johanna in every respect; but for Mintrop she was the only one, and yet she was another's.

Three years had passed since Mintrop worked his love into his art—throwing but a thin veil of grotesqueness over his real feelings; and Johanna returned from afar with her husband. They settled in Westphalia; and Johanna, moved by the memories of old days, proposed that Mintrop should be godfather to their infant daughter. Three years were gone, and Mintrop thought he had conquered his hopeless love; but yet the request startles him, and he requires to struggle for composure before he can determine whether he shall agree to it or not. He goes, finds the comfortable home where his lost love resides, meets her and her husband and the various guests present at the ceremony. The priest comes, and the little soft baby is placed in his arms. He looks at his sleeping god-daughter as he somewhat awkwardly receives her, and the child slowly opens her large eyes, so like her mother's. A thrill runs through Mintrop's veins; all the old feelings, the old hopes and fears, rush through his mind with a force too cruel to be borne. He hastily places the child in its mother's arms, and hurries away from the scene.

Not long after, and Mintrop is dying. Some physical cause, the doctor assigns; but his friends know well what it is. His patient loving heart has borne too much. The intensity of his feelings has snapped the cord of life. As his breath leaves him, he thinks of his other love, his Art, and he sighs:

‘Would I might live long enough to finish my work; otherwise, I am ready to die!’ And thus the brave gentle spirit went forth to meet its Maker, regretting only that the promise of its youth was not fulfilled—the work not yet completed. Alas, alas, for human love, for human hopes and wishes! My eyes are wet as I trace these concluding lines; and the face in the photograph is hallowed by a strange sad interest.

Theodor Mintrop died at Düsseldorf in July 1870; and his sad story, as given above, speedily found its way into the German newspapers. In autumn 1871, a bronze bust erected to his memory was unveiled in the presence of thousands of spectators; and the poet Emil Rittershaus composed and recited a beautiful poem—a requiem to one who died of a broken heart.

THE MONTH: SCIENCE AND ARTS.

THE rumour mentioned in our last *Month* has been verified, and we now know that hydrogen and nitrogen have yielded to the power of the physicist, and that there is no longer, in our part of the universe, any such thing as a permanent gas. After Pictet in Geneva had led the way by liquefying oxygen, Cailletet followed in Paris with the other two; but Pictet has since gone farther, and has obtained liquid hydrogen in considerable quantity, and has produced solid particles of oxygen. In communicating these facts to a scientific body in Paris, Mr Dumas, the eminent chemist, stated to his hearers they might take it for granted that in swallowing a glass of water they were really drinking a metallic oxide.

Dr Angus Smith says in a paper ‘On the Examination of Air,’ read before the Royal Society, that there ought to be observatories for Chemical Climatology and Meteorology, in which the air should be systematically examined, ‘so as to obtain decidedly those bodies which have from the earliest times been supposed to exist in it, bringing with them, on certain occasions, the worst results.’ But the process of examination, as at present carried on, is slow and troublesome; when a sure and easy way is found, then its adoption may become general. Dr Angus Smith is perhaps the first who has taken the subject in hand from this point of view. ‘It is the more interesting,’ he remarks, ‘as he has sufficiently shewn that in the places examined, the organic ammonia has been in intimate relation with the gross death-rate.... It may be true that oxygen is the prime mover—producing in man animal life—a favourite idea for a chemist; but it may also be true that minute organisms cause a peculiar class of decomposition connected with mental or other activity, diseased or otherwise.’

Before the telephone has ceased to be a scientific novelty, America sends us news of another novelty called a phonograph. This instrument, the invention of Mr T. A. Edison, makes sound visible, and records it in a permanent form. You speak into a tube, and while doing so you work a handle which causes a cylinder to revolve; the sound of the voice causes a thin disk or

diaphragm of metal to vibrate, as in the telephone; the vibrations actuate a steel point which, as it advances and recedes, makes impressions more or less deep in a band of tinfoil wound round the cylinder, and this band of tinfoil becomes the record of what has been spoken. Now comes the wonderful part of the process; for we are told that if the tinfoil so indented be applied to another instrument, called the 'transmitter,' consisting of a hollow tube with a paper diaphragm, then the original sounds will be reproduced, though with somewhat of a metallic tone. Turn the handle of the cylinder and you may have repetitions of the discourse until, in fact, the tinfoil is quite worn out. Casts of the indented tinfoil may, it is said, be taken in plaster of Paris, so that copies of spoken words could be sent to as many persons as may be desired.

This invention seems too questionable to allow of any one, even the inventor, forming an opinion as to its practical value. Fanciful conjectures may of course be made. A fugitive swindler, for example, may be arrested in a foreign city, and held fast until a foil of evidence spoken by one of his confederates might be sent out to convict him. Or a hardy young sheep-farmer in Australia might sing into his tube, puncturing his song on the sheet of foil, fold it neatly up, and send the graven song home to the girl he left behind him; and she, by applying the sheet to her own phonograph might, by proper manipulation, hear the tender ditty as often as she pleased.

While waiting for further developments, we venture to suggest that what is wanted by numbers of intellectual people who find the mechanical action of writing slow and irksome, is, some kind of 'graphy' which will enable them at once to print their thoughts on paper without aid from pen or fingers.

Some months ago we mentioned the little torpedo boat *Lightning*, and her swift steaming, nineteen knots an hour. Her length is eighty-four feet, her width ten feet ten inches: and now we hear that fifteen similar vessels are to be built, and that the builders promise a speed of twenty-five knots. Experiments have been made which prove that swiftness is an element of safety, for on firing a rifle-bullet through the bottom it was found that the water did not enter. In future it is thought that torpedoes will play an important part in naval warfare; and as has already been mentioned in recent papers in this *Journal*, a School has been established at Portsmouth in which their use is taught theoretically and practically. A further improvement is whispered in certain quarters—a torpedo boat which shall

carry on her evolutions under water, and hook on torpedoes to the bottom of an enemy's ship without being discovered. Are we about to see in this a realisation of what has long been a dream among speculative inventors? Is naval warfare, from its hopelessly fatal nature to those engaged, to become an impossibility?

Communications addressed to the Société d'Encouragement pour l'Industrie Nationale, Paris, describe a method for preventing the deposit of soot in chimneys; but as yet no details are published: also an apparatus for stopping runaway horses (in harness), by completely closing the winkers; and a way to deaden the blows of a hammer moved by machinery. In this case, the anvil is supported on a float in a reservoir of water. Another subject is a tramway car in which compressed air is the motive-power, as proved during some months on the line between Courbevoie and Puteaux, and the Round Point in the Champs-Élysées. This car has room for thirty passengers, is served by a conductor, and a mechanic who has entire charge of the machinery, which with a number of iron tubes is all placed between the wheels, under the floor, where it occasions no inconvenience to any one. A powerful air-pump at the starting station, forces air enough into the iron tubes for the journey to and fro, and the car travels smoothly and without noise or smoke, and can be stopped and started more readily than a horse-car. Mr Mékarski, the inventor of this car, has been thanked by the Société for having solved the problem of a locomotive which can be used with safety in crowded streets. Of course there are appliances for regulating the pressure of the air, and for preventing the deposit of hoar-frost in the tubes, consequent on rapid expansion of air; but for a description of these and other particulars we must refer to the *Bulletin* published by the Society.

Mr Coret has invented what he calls a self-acting thermo-signal which by ringing a bell makes known to all within hearing when an axle or any other part of an engine is over-heated. It is a small brass cylinder, containing a system of flexible metal disks, and a dilatable liquid, which is to be fixed to the part liable to over-heating. While all goes well the instrument makes no sign; but as the temperature rises the liquid dilates, forces out a small metal pin at the end of the cylinder, which, as the wheel revolves, strikes a bell, and thereby warns the attendants. Thus the necessity for constantly watching an indicator is avoided.

Other subjects brought before the same Society are—A description of a chimney which does not occasion loss of heat, by Mr Toulet, 38 Avenue de l'Observatoire, Paris—Specimens of harmless colours which may be used with varnish, oil, or water, and are described as durable and remarkably brilliant. They are available for many purposes of decoration, but are specially intended, as they contain no poisonous element, for the colouring of children's toys. These new colours are derived from the substances known to chemists as eosin and fluorescin—And certain manufacturers who have carefully studied the material give an account of the capabilities of jute, from which we gather that by proper preparation of the yarns, remarkable effects of colour, of mottling, of light and shade, and also a velvety appearance can be produced. The process is described as very simple and moderate in cost; so that applications of jute to decorative purposes hitherto not thought of may ere long become available.

It has been found by experiment that aniline black can be made to yield different colours: treated in one way it is a light violet, in another way it is a bluish pink, and in a third way it becomes blue.

Pure butter, as is stated in the *Journal* of the Chemical Society, contains from ninety to ninety-eight per cent. of pure butter fat and a small quantity of water. Its colour should be from yellowish white to reddish yellow, but this depends on the kind of fodder given to the cows, and may be produced by means of beetroot or other plants possessed of colouring properties. The colouring matter may be detected by treating the butter with strong alcohol. The melting-point of pure butter is from thirty to thirty-seven degrees, while artificial butter melts at from twenty-seven to thirty-one degrees. Substances used to increase the bulk and weight of butter are chalk, gypsum, oxide of zinc, starch, and so forth. These neither improve its flavour nor its wholesomeness. The agreeable smell of pure butter, with a slight suggestion of milk, is not easy to imitate by artificial means.

Now that chemists can avail themselves of the spectroscope in their researches, falsifications have but little chance of escaping detection. We learn from the same *Journal* that the colouring matters generally used in the adulteration of wine are—fuchsine, the preparations termed caramels, ammoniacal cochineal, sulphindigotic acid, logwood, the lichen reds, rosaniline, bilberries, cherries, mallows, and the berries of the privet. Most if not all of these matters can be precipitated by chemical treatment, or they

may be detected by dialysis. If a cube of gelatine less than an inch square be placed in the wine under experiment, it will be found, after twenty-four or forty-eight hours, stained all through, if artificial colouring matters are present; but if the wine is quite pure, then the natural colouring matter will not have penetrated deeper into the gelatine than one-eighth of an inch. It is worth notice that the natural colour soaks in slowly; the artificial colour quickly.

The *Comptes Rendus* of the Académie des Sciences, Paris, give an account of a patient who, through entire closure of the esophagus or gullet, could get neither food nor liquid into his stomach, and had to undergo the operation of gastrotomy. Through the opening thus made the operator passed different substances and took note of the time they remained in the stomach. Starch, fat, and flesh disappear in from three to four hours; milk is digested in an hour and a half or two hours, and alcohol and water are absorbed in from thirty-five to forty-five minutes. One day a small quantity of pure gastric juice was taken from the stomach for experiment: it is described as colourless, viscid, yet easily filterable, having little odour, and not putrefying spontaneously. The acidity of the gastric juice varies but slightly whether mixed with food or not, the mean being 1.7 gram of hydrochloric acid to one thousand grams of liquid. 'The quantity of liquid,' we are informed, 'found in the stomach has no influence on its acidity; the latter is almost invariable whether the stomach be nearly empty or very full. Wine and alcohol increase the acidity, while cane-sugar diminishes it. If acid or alkaline liquids are injected into the stomach, the gastric juice reassumes its normal acidity in about one hour. It is more acid during digestion than when digestion is not going on, and the acidity increases towards the end of the process. Since the stomach is generally empty at the end of four hours, and hunger does not supervene till about six hours after a meal, it would seem that hunger does not result solely from emptiness of the stomach.' This last remark is not in accordance with the opinions of other physiologists; but we venture to suggest that in common with the limbs, the stomach needs rest, and finds it in the two hours of quiet above mentioned. We would further remark, that the theory that sugar does not create acid in the stomach is contrary to all ordinary medical teaching, and even of daily experience.

A surgeon in a provincial town in Scotland has achieved a remarkable operation. He cut out from the neck of a patient a diseased portion of the

larynx, and inserted an artificial larynx through which the man can speak articulately. This is one of the triumphs of surgery.

We mentioned some time ago that certain practitioners in the United States had succeeded in removing tumours by the application of a current of electricity. Recently the same method has been employed, and with the same success, for the removal of those blemishes from the skin popularly described as ‘port-wine stains,’ and other excrescences. Care is required in regulating the strength and duration of the current according to the nature of the case; if this be insured, the operation can hardly fail of a successful result. Particulars of cases and their treatment are published in the *New York Medical Journal*.

Pursuing his contributions to meteorology, Professor Loomis of Yale College, Newhaven, U.S., finds that the areas of rainfall in the United States generally assume an oval form, and the oval is not unfrequently a thousand miles long and five hundred broad. He finds too that falls of rain often have great influence in checking the progress of a storm; and that they appear to be subject to some law of duration. For example, some rains last eight hours, some sixteen, some twenty-four; but beyond twenty-four hours the instances are very rare. ‘This fact,’ he remarks, ‘seems to indicate that the causes which produce rain, instead of deriving increased force from the rainfall, rapidly expend themselves and become exhausted. It cannot be explained by supposing that the vapour of the air has all been precipitated, because these cases chiefly occur near the Atlantic coast, where the supply of vapour is inexhaustible. Is there not here an indication that the forces which impart that movement to the air which is requisite to a precipitation of its vapour, become exhausted after a few hours’ exercise?’ By further research it is found that during the six months from November to April, violent winds are more than five times as frequent as during the other six months of the year; and that they come from a northern quarter two-and-a-half times more frequently than from a southern quarter. Though Professor Loomis’ observations apply to the climate of America, they may be considered with advantage by our own meteorologists.

The President of the Institution of Civil Engineers in his inaugural address took occasion to say, as evidence of the advantages which accrue to a country through the labours of the civil engineer, that the sum authorised to be expended on British railways up to the end of 1876 amounted to seven

hundred and forty-two millions; a sum pretty nearly as large as our huge national debt. And from this Mr J. F. Bateman argued, that as in engineering special qualifications, and some of a high order, were required, it would be well if advantage were taken of the numerous public schools in which instruction bearing on engineering is given, whereby young men would have at least some qualification on entering the profession. At the same time it would be a mistake to regard that training as other than preparatory and incomplete. It is by actual outdoor work only, that a man can become an engineer; and engineering work is not to be found at school or college.

Mr Bateman—who by the way will long be remembered for his water-supply of Glasgow—instead of travelling over many topics, confined himself to the great and important question of rainfall and water-supply for the whole kingdom, with a view to proper economy. It is a question which becomes more and more important with the increase of population and consequent multiplication of machinery. When the Metropolitan Board of Works are about to ask parliament for leave to undertake the water-supply of London, the proportions of the question may be assumed to be at their largest; and storage of rainfall and of flood-waters, prevention of pollution, and the best way of obtaining absolutely pure water, together with other topics, will have to be treated with serious consideration.

SPRING.

OFT let me wander hand in hand with Thought
In woodland paths and lone sequestered shades,
What time the sunny banks and mossy glades
With dewy wreaths of early violets wrought,
Into the air their fragrant incense fling,
To greet the triumph of the youthful Spring.
Lo! where she comes! 'scaped from the icy lair
Of hoary Winter; wanton free and fair!
Now smile the heavens again upon the earth,
Bright hill and bosky dell resound with mirth,
And voices full of laughter and wild glee
Shout through the air, pregnant with harmony,
And wake poor sobbing Echo, who replies
With sleeping voice that softly, slowly dies.

ERRATUM.

[The verses which appeared in last month's issue, entitled *The Well-known Spot*, were signed by mistake ASTLEY H. BALDWIN instead of F. G. ELLIOTT. We take this opportunity of rectifying the error.]

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